

C7 CIVIC STATION | One 1:1 city-completeness prototype

ordinary city → staffed service → optional AI → controlled test → AI-OFF recovery

Ordinary-city base

- 01 Shade + seating
Stay first; interact later
- 02 Continuous access
AI cannot patch a physical barrier
- 03 Fixed bilingual wayfinding
Arrival still works offline

Staffed + optional AI

- 04 No-account counter
Paper/phone/human same task
- 05 Optional AI
Minimal data; refusal allowed
- 06 Human takeover
High-risk or failure goes human

Controlled test + recovery

- 07 Test pocket
Physically separate from passage
- 08 Immediate stop
Stop right before scaling
- 09 AI-OFF
Basic city function remains

Concept prototype only; no site approval, construction, field performance or engineering clearance is claimed.

One prototype, three urban embeddings

Fixed reviewer input: C7 CIVIC STATION is embedded differently in three urban fabrics, not repeated as generic task cards.

TEST

Zhongzhiyuan | innovation campus

R&D arrival, meals, rest and ordinary walking stay continuous; controlled robot/device tests occupy a separate pocket.

CARE

AI Origin | long-term community

Home, shade, seating, staffed care/service and commons work first; no phone or account is required for the basic task.

ARRIVE

Dazhongsi | station-city life

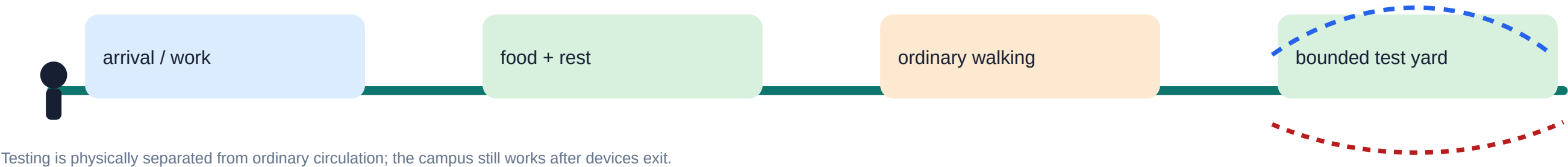
Fixed bilingual wayfinding, staffed interchange, ordinary commerce and rest work first; dynamic AI falls back to physical routes.

All key-area geometry remains provisional; this shows spatial roles, not station exits, parcels or engineering positions.

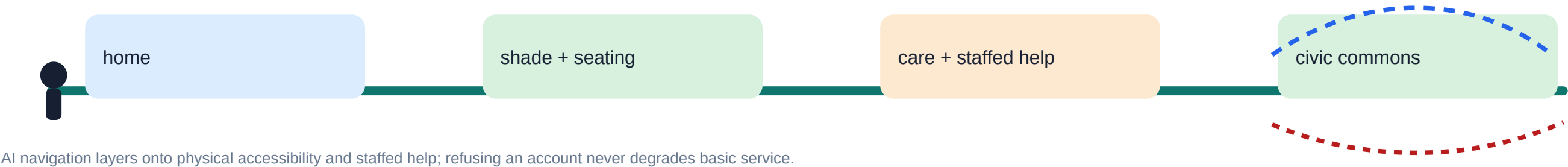
v0.7 | Three everyday journeys: start with a day in the city, then place AI

Black=people/ordinary city · Green=physical repair · Blue dash=optional AI · Red dash=functioning exit path

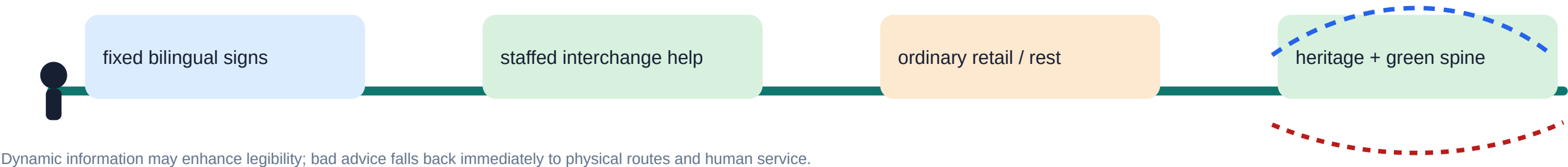
01 Zhongzhiyuan | researcher + service worker



02 AI Origin | older resident + carer + no-phone user



03 Dazhongsi | commuter + international visitor + service worker

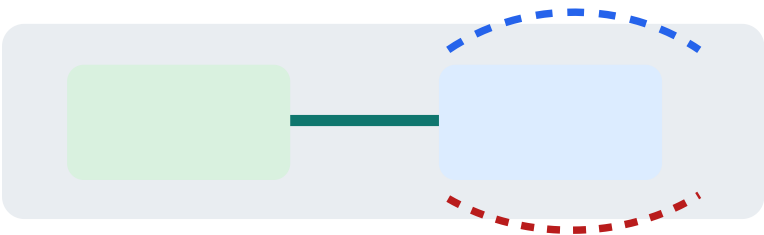


Concept-only · provisional geometry · real entrances, roads, permits, insurance and accountable parties require verification.

v0.7 | How AI changes urban form: six reversible physical interfaces

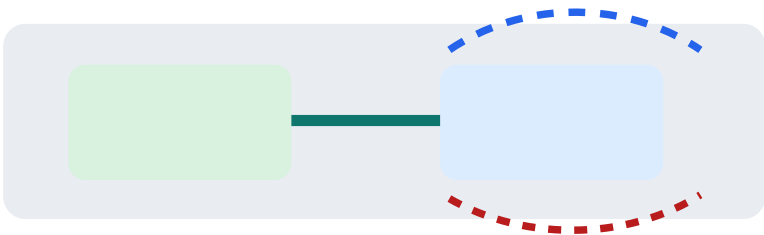
ordinary-city baseline → physical repair → optional AI → functioning exit state

A | Test pockets



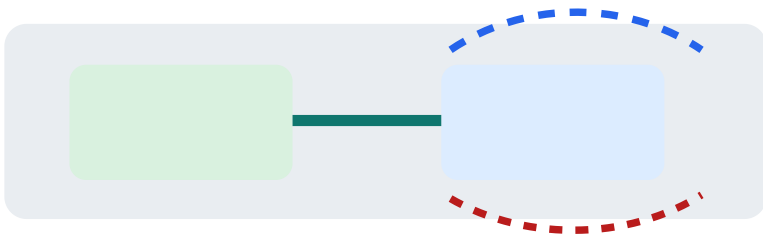
Separate testing from ordinary circulation; remove devices and keep
AI is optional; physical city function remains after exit.

B | Accessible help nodes



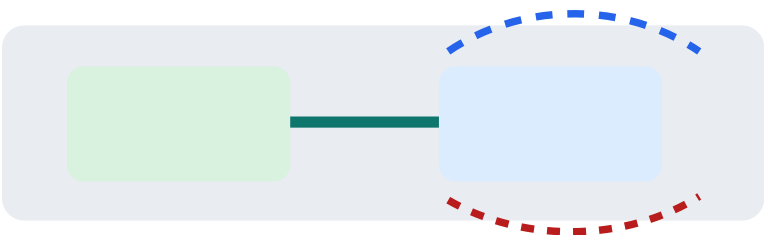
Seating, shade, static signs and staffed help come first; navigation/tr
AI is optional; physical city function remains after exit.

C | Continuous station-city arrival



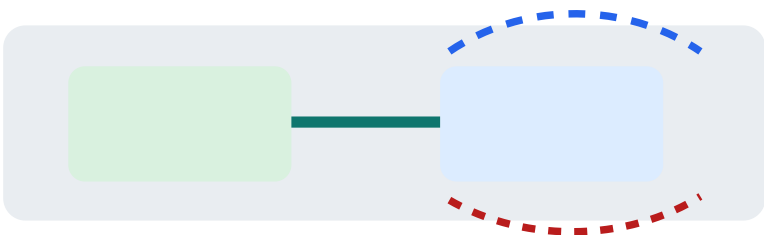
Fixed bilingual signs, staffed help, rest and ordinary retail form the basi
AI is optional; physical city function remains after exit.

D | Replaceable service nodes



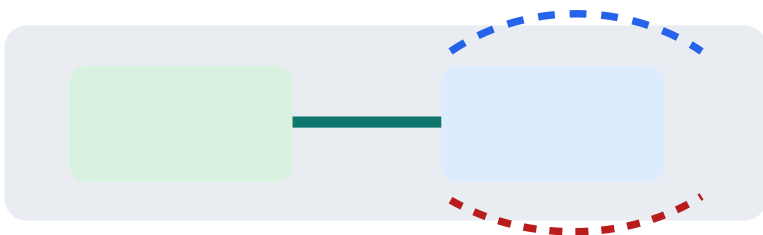
Edge compute, charging and maintenance become removable nodes
AI is optional; physical city function remains after exit.

E | People-first public ground floors



Space for sitting, staying, food, meeting and staffed service takes pri
AI is optional; physical city function remains after exit.

F | Reversible spatial versioning



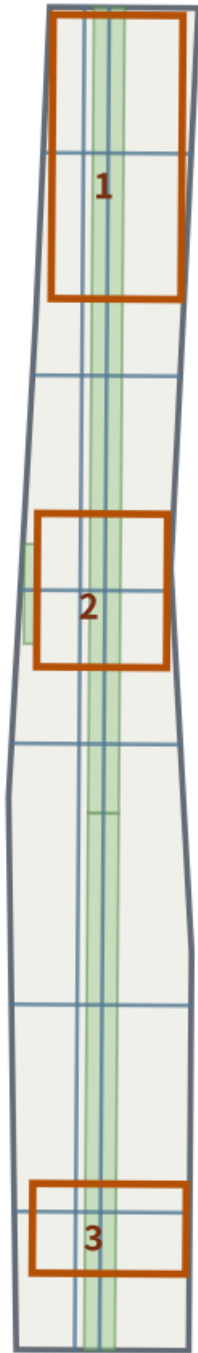
Observe → bounded prototype → public/professional review → merge
AI is optional; physical city function remains after exit.

Shared principle: AI is not an eighth land-use category; it creates spatial interfaces that can start, stop and be replaced.

Concept-only • all dimensions and engineering interfaces require later verification.

01 | Site Overview + City Completeness

One civic green spine • six completeness segments • six east-west stitches • three key-area cores



Three-scale work frame

- Coordinated research ~43.6 km² | ecosystem + talent + future city
- Overall design ~11.4 km² | continuous spatial structure
- Three key areas ~368.4 ha | detailed design tasks

C7

HOME • LEARN • CARE • MOVE • GREEN • WORK • COMMON LIFE

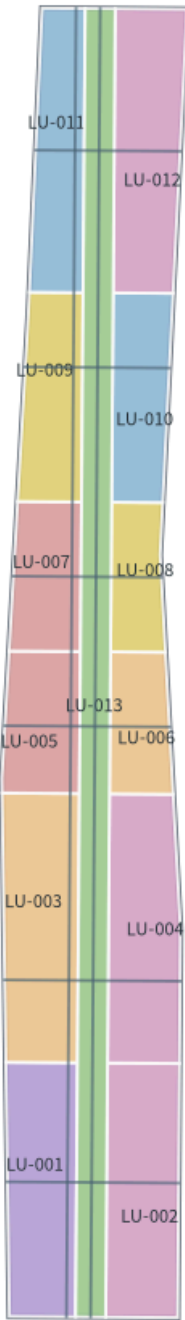
AI is an optional enhancement across C7, not an eighth land-use category.

Concept proposal • provisional geometry is not an official redline • recalculate when official data arrives

Site overview and C7 method

02 | C7 Land-use + Civic Base Structure

13 conceptual land-use features; allowed codes only; AI does not create a new statutory land-use class



Legend

- Residential 0701
- Community service 0702
- Research 0802
- Culture 0803
- Education 0804
- Commercial service 05
- Park green 1401

Design discipline

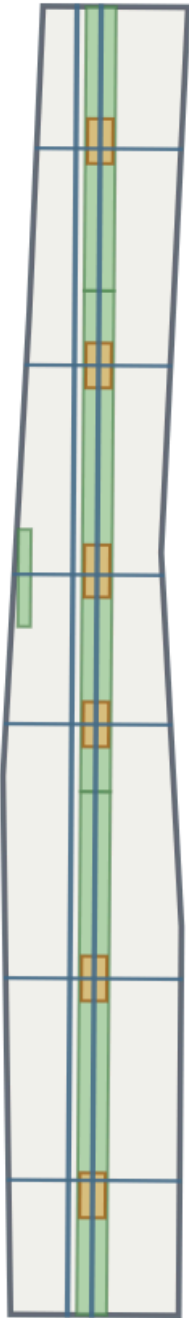
- Concept area ≠ statutory zoning
- conceptual_floors ≠ approved height
- Six segments are work zones, not administrative communities

Concept proposal • provisional geometry is not an official redline • recalculate when official data arrives

Land-use structure

04 | Mobility • Blue-Green • Civic Space

The civic green spine first serves walking, cycling, accessibility, shade, stormwater and free staying



Public-first rules

- No account / no app required
- AI guidance is optional
- Six E-W links are intentions, not road redlines
- Transit exits, structures and parking await professional data

- 绿地 / Green
- 公共空间 / Civic space
- 慢行联系 / Slow mobility

Concept proposal • provisional geometry is not an official redline • recalculate when official data arrives

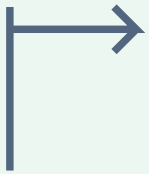
Mobility and blue-green system

Jing-Zhang City Completeness v0.5 | Public-Space Component Library + Three-Layer Wayfinding

Design order: permanently usable ordinary city function → updateable operational information → optional AI enhancement. If the digital layer fails, the space remains understandable and usable.

Seven reusable components

01 C7 ORIENTATION POST



Permanent: direction / distance / C7 tags

equal-rank CN/EN; icon + text; no QR dependency

Digital enhancement

optional voice/multilingual Q&A; offline backup

02 SHADED STAY ISLAND



GREEN + COMMON LIFE

seating, shade, water/charging resource

Digital enhancement

optional environment info without id

03 ROBOT HANDOFF BAY



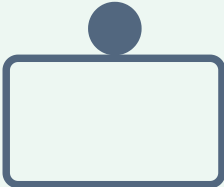
MOVE + WORK

robots stay out of ordinary paths; physical

Safety rule

low speed, closable, human takeover, b

04 STAFFED SERVICE DESK



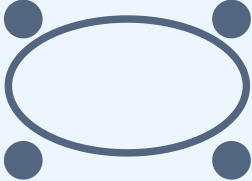
CARE + NON-AI BASELINE

booking, appeal, navigation, and help all have a human

Spatial requirement

accessible counter, clear hours, emergency contact

05 COMMUNITY SHARED TABLE



COMMON LIFE

free core hours, public booking rules

AI scheduling

advisory only; fairness/conflicts can

06 ACCESSIBILITY CONTINUITY NOT



MOVE + CARE

ramps/rest/tactile/light exist before a

temporary-obstacle AI is an extra al

07 JING-ZHANG MEMORY RAIL



CULTURE + LEARN

dates, people, engineering, commu

Multilingual layer

fixed CN/EN + optional multilingual

Shared component rules

- Real daily function first; communication/AI second.
- Screens never replace seating, lighting, ramps, or physical signs.
- Devices never intrude into fire access, tactile paths, circulation, mainten
- C7 tags are common; materials may differ by key area.

Final dimensions, structure, materials, and devices need professional design.

Three-layer wayfinding: start with “still works when power is off”

L1 PHYSICAL | ALWAYS-ON

direction, distance, accessibility, public service, hazard boundary, fixed heritage te

equal-rank CN/EN, icon+text, color not sole encoding, night readability

failure mode: movement and help still work with no power/network/phone

Owner: spatial/facility operator

L2 OPERATIONS | CHANGEABLE

hours, temporary closures, events, test status, queues, restoration info

every item has update time + responsible role; stale info is removed

failure mode: uncertainty/emergency switches to staffed notices

Owner: information updater + on-site human review

L3 OPTIONAL AI | OPT-IN

multilingual Q&A, personalized routes, scenario explanation, booking assistance

traceable source, minimal data, human takeover, exit recovery

failure mode: shut AI and fall back to L1/L2 with no loss of basic service

Owner: AI service role + corresponding public-service operator

International communication syntax

Which ordinary city capability does it support? → What does AI add? → How do I use it without AI? → Who is responsible?”

Each public node answers these five questions in Chinese and English, replacing “future-tech” slogans with function, rights, and responsibility.

Concept component library only; not construction drawings, a procurement list, facility permit, or official wayfinding standard.

v0.6 | Three flagship pilots: validate before scaling

All quantities and durations are conceptual test configurations, not approved budgets, contracts, permits, sites, or partnership commitments.

01 Zhongzhiyuan | Low-speed robots

Question: how can R&D testing avoid turning public routes into default test spaces?

NON-AI BASELINE

Ordinary walking + human delivery + staffed on-site guidance

Concept test configuration

1 bounded / controllable yard; 2–4 low-speed devices; at least 2 human marshal roles per active test window

Prerequisite evidence gates

Site permission, fire/egress, device safety, insurance/liability, physical separation and human takeover confirmed before start

Window / decision

14–30 controlled days → GO / REVISE / STOP

KPI

No spill into unauthorized routes; takeover reachable; ordinary movement not degraded; incidents reviewable

Automatic stop threshold

Boundary breach, loss of link, blocked accessible path, takeover failed

Exit receipt

Device removal + temporary-space reset + incident/data cleanup review

Responsibility structure

A: real operating authority (to be confirmed)
R: test lead; C: safety/accessibility/maintenance;
I: public and site management

02 AI Origin | Access / care navigation

Question: can digital assistance enhance, rather than replace, physical accessibility?

NON-AI BASELINE

Continuous physical accessibility + fixed signs + staffed help desk

Concept test configuration

1 route already checked physically; 10–20 voluntary, withdrawable use sessions as a concept cap

Prerequisite evidence gates

Accessibility review, service hours, human help, data minimization, consent/exit and equal service without registration confirmed

Window / decision

2–4 week small-scale co-test → GO / REVISE / STOP

KPI

Route completion not below baseline; human help reachable; refusing tracking never removes basic service; errors appealable

Automatic stop threshold

Misdirection to inaccessible path, sensitive-data breach, no human help

Exit receipt

Personalization off + deletion/migration record + baseline continues

Responsibility structure

A: real public-service authority (to be confirmed)
R: staffed service lead; C: accessibility/privacy/community;
I: residents, carers and users

03 Dazhongsi | Interchange / multilingual wayfinding

Question: can AI improve legibility without creating a dependency?

NON-AI BASELINE

Static signs + staffed advice + routes understandable without a phone

Concept test configuration

1 field-verified arrival portal; 2–3 candidate routes; Chinese / English and no-phone modes all remain available

Prerequisite evidence gates

Real entrances, opening hours, property/operations interface, accessibility and emergency wayfinding verified before testing

Window / decision

14-day bounded validation → GO / REVISE / STOP

KPI

Wayfinding time/help requests improve against baseline; static signs stand alone; bad advice is human-correctable

Automatic stop threshold

Unsafe routing, conflict with live site state, failed human update

Exit receipt

AI layer off + physical signs retained + correction/closure notice

Responsibility structure

A: real station-city/public-interface authority (to be confirmed)
R: on-site information lead; C: transport/accessibility/operations;
I: commuters, visitors and nearby residents

Shared resource and cost boundary

v0.6 states quantity bases, staff roles and test durations only. Currency budget, total FTE, procurement, insurance, contracts and permits stay UNKNOWN until site and accountable parties are confirmed.

Concept proposal only. Provisional geometry remains non-official; no pilot above is represented as executed, funded, approved or contracted.

v0.6 | Implementation resources & RACI: make feasibility auditable

This table states conceptual quantity bases, roles and maintenance cadence only. Currency budgets, total FTE, procurement entities, insurance contracts and approvals remain UNKNOWN.

PILOT	RESOURCE / QUANTITY BASIS	RACI RESPONSIBILITY	MAINTENANCE CADENCE	PRE-START EVIDENCE GATES
Zhongzhiyuan Controlled low-speed robots 14–30 day concept window	• 1 controllable test yard • 2–4 low-speed devices • ≥2 human marshal roles / active window • removable separation / stop signage Cost class: M (concept) currency UNKNOWN	A: real operating authority (TBC) R: on-site test lead C: safety / accessibility / maintenance / insurance I: site management / public	Device + route check before each day Incident log every test window Daily close-down reset Weekly GO/REVISE/STOP review	Site permission Fire / egress Device safety Liability / insurance Separation + human takeover
AI Origin Accessibility / care navigation 2–4 week concept window	• 1 route already physically audited • 10–20 voluntary, withdrawable sessions cap • 1 fixed staffed help point / service window • L1/L2 non-AI wayfinding remains live Cost class: S–M (concept) currency UNKNOWN	A: real public-service authority (TBC) R: staffed service lead C: accessibility / privacy / community / care I: residents / carers / user representatives	Physical route check before daily operation Human review for every wrong suggestion Weekly retention/deletion audit Issue list published at phase close	Accessibility review Staffed service hours Data-minimization plan Consent / exit Equal no-account service
Dazhongsi Interchange / multilingual wayfinding 14-day concept window	• 1 field-verified arrival portal • 2–3 candidate routes • Chinese / English / no-phone entry • static signs retained and standalone Cost class: S–M (concept) currency UNKNOWN	A: real station-city/public-interface authority (TBC) R: on-site information lead C: transport / accessibility / property / operations I: commuters / visitors / nearby residents	State verification before daily opening Human update whenever site state changes Unsafe guidance closed/corrected same day Compare baseline on days 7 / 14	Real entrances / opening hours Property / operations interface Accessible route verified Emergency guidance consistent Human update owner named

Shared decision gate: PRECONDITION → TEST → RECEIPT → DECISION

1 Preconditions evidenced
not complete = do not start

2 Time-bounded controlled test
AI-off baseline runs in parallel

3 Exit / fault receipt
remove, correct, delete, reset

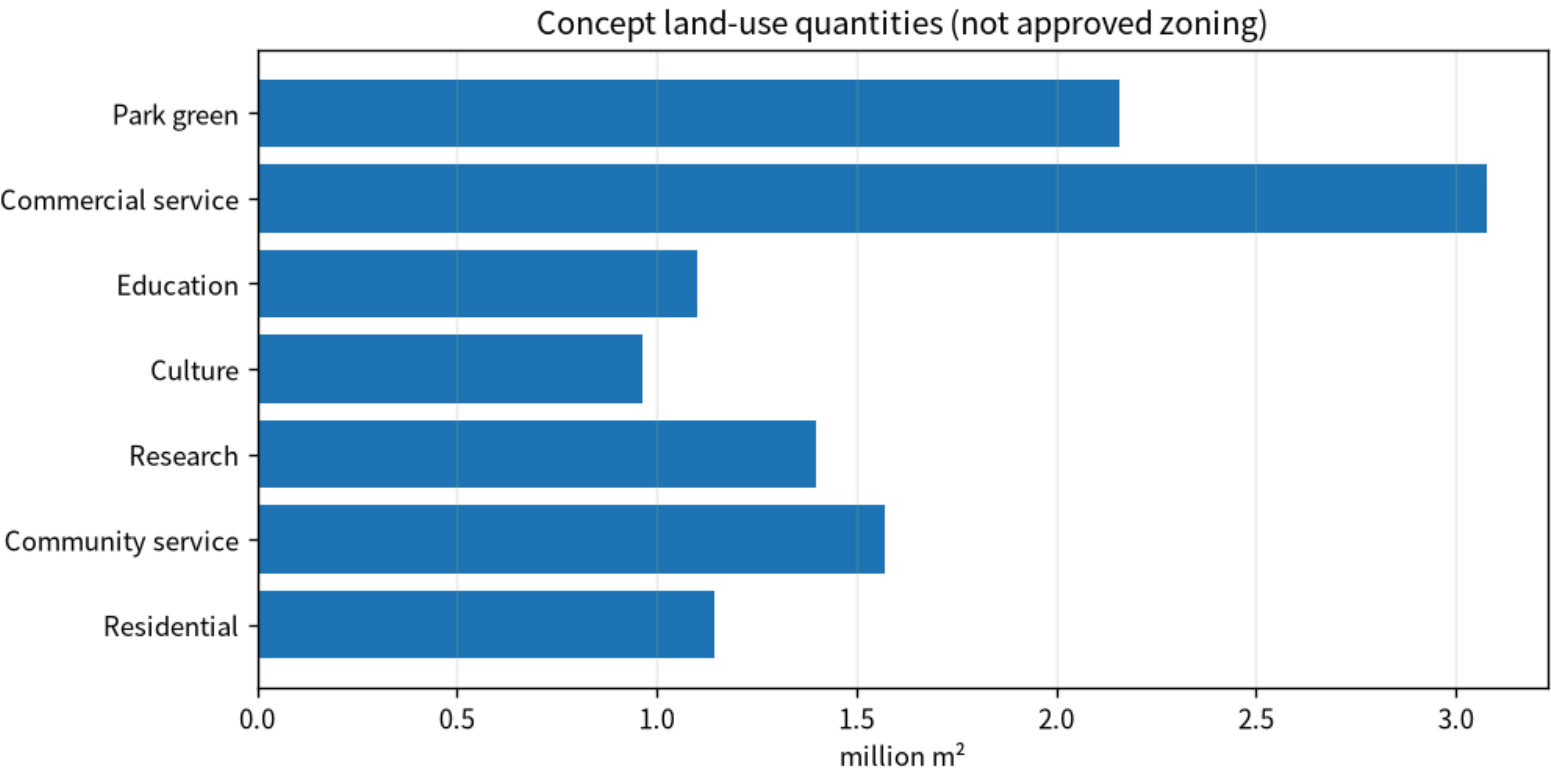
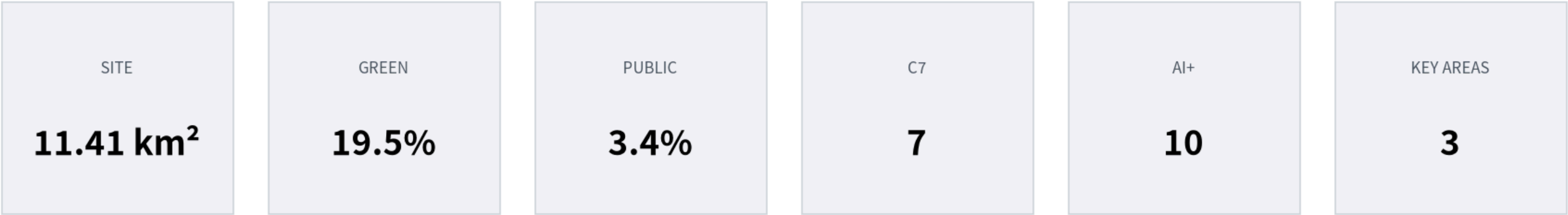
4 GO / REVISE / STOP
scaling is never the default

Shared boundary: if accountable parties, permits, safety, insurance, service hours or field conditions cannot be verified, the proposal remains a concept protocol; no fake budget, partnership or schedule fills the gap.

Concept-only implementation framework. Provisional geometry is non-official and is not a procurement, approval, contract or construction basis.

05 | Metrics • Evidence • Data Gaps

Known values, conceptual design quantities and unknown statutory/engineering quantities are kept separate



Critical data gaps

- Official precise site + three key-area polygons
- Official FAR / height / road redlines / heritage / fire / utilities
- Cleared building, tenure, household and service-capacity survey
- Verified road widths, parking, transit interfaces and engineering constraints

Recalculation trigger: when official geometry or key controls arrive, recompute land use, buildings, roads, green/civic space, phasing, metrics, drawings and HTML.

Concept proposal • provisional geometry is not an official redline • recalculate when official data arrives

Metrics and evidence limits